Backpaper Exam - Discrete Mathematics I
B. Math II

29 December, 2025

(i) Duration of the exam is 3 hours.

(ii) The maximum number of points you can score in the exam is 100.

(iii) You are not allowed to consult any notes or external sources for the exam.

Name: ____________________

Roll Number: ____________________

1. (15 points) For a positive integer $n \geq 2$, let $\phi(n)$ denote the number of positive integers in $\{1, 2, \ldots, n\}$ which are coprime to n. If $p_1, \ldots, p_k$ are the distinct prime factors of n, prove that
$$\phi(n) = n \prod_{i=1}^{k} \left(1 - \frac{1}{p_i}\right).$$

Total for Question 1: 15

2. (20 points) Assuming that there are sufficiently many beads of colours *red* and *blue*, find the number of distinct necklaces with n beads that can be constructed.

Total for Question 2: 20

3. (15 points) Let $G = (V, E)$ be a connected finite simple graph. Prove that G has a path of length at least $\delta(G)$, where $\delta(G) := \min_{v \in V} \deg(v)$ is the minimum degree of G.

Total for Question 3: 15

4. (20 points) Let G be a finite k-regular bipartite graph (that is, every vertex has degree k). Prove that G contains a perfect matching.

Total for Question 4: 20

5. Let $\mathcal{F} \subseteq \mathcal{P}([n])$ be an antichain.

   (a) (10 points) Prove the Lubell–Yamamoto–Meshalkin (LYM) inequality:

$$\sum_{A \in \mathcal{F}} \frac{1}{\binom{n}{\#(A)}} \leq 1.$$

   (b) (10 points) Using the LYM inequality (or otherwise), show that

$$\#(\mathcal{F}) \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}.$$

Total for Question 5: 20

6. Let $n \geq 2$ be an integer. Two Latin squares of order n are said to be *orthogonal* if, when superimposed, every ordered pair of symbols occurs exactly once. A collection of Latin squares is said to be *mutually orthogonal* if every distinct pair in the collection is orthogonal.

   (a) (10 points) Show that the number t of mutually orthogonal Latin squares (MOLS) of order n satisfies

$$t \leq n - 1.$$

   (b) (10 points) Let q be a prime power. Using the finite field $\mathbb{F}_q$, construct explicitly $q-1$ MOLS of order q (with justification), thereby showing that the upper bound is attained in this case. (Hint: You may define $L_a(i,j) = ai + j \pmod q$ for $a \in \mathbb{F}_q^*$.)

Total for Question 6: 20